

Dataset Description

IBM HR Analytics Employee Attrition & Performance

1. Source and Domain

The dataset is a fictional but realistic Human Resources dataset originally published by IBM data scientists as part of the IBM Watson Analytics sample data collection, and widely distributed on Kaggle as “IBM HR Analytics Employee Attrition & Performance.” It falls squarely in the **Human Resources (HR)** domain listed as an approved project category.

2. Structure

The raw file contains **1,470 employee records** and **35 columns**, comfortably exceeding the minimum requirement of 1,000 rows and 8 variables. Each row represents one employee; columns capture demographics, compensation, job characteristics, satisfaction survey results, and tenure history.

3. Target Variable

Attrition (Yes / No) — whether the employee has left the company. This is a clean, clearly defined binary classification target with an approximate 84% / 16% class split, making it well suited for supervised machine learning (classification) while still presenting a realistic class-imbalance challenge to handle.

4. Variable Types

Category	Examples	Count (approx.)
Numerical (continuous/discrete)	Age, MonthlyIncome, DistanceFromHome, TotalWorkingYears, YearsAtCompany	~26
Categorical (nominal/ordinal)	Department, JobRole, BusinessTravel, MaritalStatus, EducationField, Gender	~9
Target	Attrition (Yes/No)	1

5. Data Quality (as sourced vs. as used)

The original Kaggle release of this dataset is unusually clean (no missing values, no duplicates). To meet the project's requirement to demonstrate real-world data cleaning skills, a controlled, fully documented set of imperfections was introduced on top of the authentic records before the cleaning stage of the analysis. This is disclosed transparently rather than hidden:

- Missing values injected into 4 columns (MonthlyIncome, EnvironmentSatisfaction, BusinessTravel, NumCompaniesWorked).
- 15 exact duplicate rows appended.

- Inconsistent text casing/whitespace in Gender, Department, and OverTime.
- Age stored as text with a “ yrs” suffix for a subset of rows (incorrect data type).
- Outliers: a few implausible MonthlyIncome values and sign-flipped (negative) DistanceFromHome values.

All of these issues are identified and resolved in Section 2 of the accompanying Jupyter notebook, restoring the dataset to its authentic 1,470-row, duplicate-free state.

6. Why This Dataset Was Selected

- **Real business problem:** employee attrition is a costly, high-visibility HR problem — replacing an employee typically costs 50–200% of their annual salary in recruiting, onboarding, and lost productivity.
- **Rich, mixed feature set:** demographic, financial, and survey-based variables allow meaningful feature engineering (e.g., income-per-year-worked, promotion gap, composite satisfaction index).
- **Well-defined, appropriate target:** Attrition is binary, unambiguous, and directly actionable — HR can intervene on employees flagged as high-risk.
- **Analytical depth:** supports EDA, classification modeling, and dashboarding, satisfying every workflow stage required by this project.

7. Business Problem

Can we predict, ahead of time, which employees are at elevated risk of leaving — and which factors (compensation, overtime, tenure, satisfaction, career progression) most strongly drive that risk — so that HR can prioritize retention efforts where they will have the most impact?